



DIGITAL LEARNING CONTENT MANUAL

GRADE NINE

**ENGLISH
MATHS
BIOLOGY
CHEMISTRY
PHYSICS**

Table of Contents

1. Introduction	2
2. Available digital content from Grade KG- Grade 12	3
3. How to read video codes	4
4. Grade 9 Digital Content Alignment with SLOs	5
a. English	5
b. Maths.....	10
c. Biology.....	33
d. Chemistry.....	41
e. Physics	49
5. How to Use This Document in Class	55
6. Ethical and Copyright Considerations.....	56
7. Conclusion.....	56

Digital Content Practical Lookup Guide

1. Introduction

The Ministry of Federal Education and Professional Training is dedicated to enhancing the quality of distance learning and creating a comprehensive educational platform that serves a diverse group of learners across Pakistan. As part of this effort, the Ministry has procured digital content, including audio and video resources, that align with the Pakistan National Curriculum and Student Learning Outcomes (SLOs).

This guide is designed to help teachers navigate this digital content efficiently. Every video has been assigned a unique code, and this document organizes them in a way that clearly shows which video supports which SLO, for which grade and subject. Each table includes the video code, topic, SLO(s) covered, textbook unit, and duration.

Teachers can use this guide for lesson planning, in-class instruction, or revision sessions. It ensures that digital resources are used effectively, keeping lessons aligned with the curriculum and maximizing learning outcomes. The structure is consistent across all grades and subjects, making it easy to locate the right video at the right time.

This guide is intended as a practical companion to support teachers in delivering high-quality, curriculum-aligned instruction through digital content.

2. Available digital content from Grade KG- Grade 12

Digital Learning Content (Video)

Grade	Subject	No. of Packages	Total No. of Packages
KG	Math	21	133
	English	36	
	Urdu	25	
	General Knowledge	51	
Grade 1	Math	35	147
	English	33	
	Urdu	60	
	General Knowledge	19	
Grade 2	Math	45	166
	English	71	
	Urdu	16	
	General Knowledge	34	
Grade 3	English	92	176
	General Knowledge	29	
	Maths	39	
	Urdu	16	
Grade 4	English	92	190
	Math	36	
	Science	42	
	Urdu	20	
Grade 5	English	73	155
	Math	23	
	Science	34	
	Urdu	25	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 6	Maths	31	120
	Science	27	
	English	37	
	Computer Science	25	
Grade 7	English	64	167
	Maths	51	
	Science	24	
	Computer Science	28	
Grade 8	English	60	162
	Maths	45	
	Science	25	
	Computer Science	32	

Grade	Subject	No. of Packages	Total No. of Packages
Grade 9	Biology	19	119
	Chemistry	21	
	English	20	
	Maths	42	
Grade 10	Physics	17	131
	Biology	26	
	Chemistry	12	
	English	21	
Grade 11	Maths	55	552
	Physics	17	
	Biology	110	
	Chemistry	94	
Grade 12	English	55	544
	Maths	67	
	Pak. Studies	45	
	Physics	56	
ECD - Grade 12	Urdu	132	40
	Gamified Educational Content	40	

Total no. of Video Packages	2802
------------------------------------	-------------

Sesame, Idara Taleem o Agahi (ITA), Environment & Entrepreneurship

Sesame Workshop	
Social Emotional Learning (SEL)	17
Maths	38
Science/ General Knowledge	21
Total	76 videos 55 (5 min videos) and 21 (27 minute videos)

ITA - Storytelling	
Grade 1	6
Grade 2	6
Grade 3	7
Total	19

Theme	No. of Packages
Environments for Living Lot 1 (ECD-2)	6
Environments for Living Lot 2 (Grades 3-5)	18
Environments for Living Lot 3 (Grades 6-8)	21
Entrepreneurship (Grades 6-8)	20

3. How to read video codes

Every video in this digital content collection has a unique code. The code helps teachers quickly identify:

- the grade
- the subject
- the textbook unit
- the package and part of the video

Understanding the code allows you to find the right video quickly without searching through multiple files.

Video Code Format

Example: **G01EnU01P01a**

- G01 = Grade 1
- En = English
- U01 = Unit 1
- P01a = Package 1, Part a

Code Reference Table:

Grade	Subject	Unit	Package
G01-G12, EY	M (Mathematics)	U01	P01
G01-G12, EY	Ur (Urdu)	U01	P01
G01-G12, EY	En (English)	U01	P01
G01-G12, EY	Sc (Science)	U01	P01
G01-G12, EY	Ph (Physics)	U01	P01
G01-G12, EY	Bi (Biology)	U01	P01
G01-G12, EY	Ch (Chemistry)	U01	P01
G12	PS (Pakistan Studies)	U01	P01

Notes for Teachers:

- Use this table to decode video codes quickly.
- “EY” refers to Early Years/Kindergarten.
- Package numbers indicate video sequence within a unit.
- Always cross-check the mapping tables in this guide for SLO coverage before showing a video in class.

4. Grade 9 Digital Content Alignment with SLOs

a. English

Unit	SLO Code	SLO	Package Code	Package Name
The Superheroes Amongst Us	1.2	Apply critical thinking to interact with text and use intensive reading strategies (while-reading) to • scan to answer short questions. • make simple inferences using context of the text and prior knowledge. • distinguish between what is clearly stated and what is implied . • deduce meaning of difficult words from context. • use context to infer missing words. • read silently with comprehension and extract main idea and supporting detail. • scan to locate an opinion. • distinguish fact from opinion in letters to editors, texts supporting an opinion, etc. • locate examples to support an opinion e.g. a personal anecdote, quotations, examples and analogy, etc. • recognize arguments and counter arguments. • explore viewpoints/ ideas and issues. • follow instructions in maps or user instruction manuals and forms related to extended social and academic environment. • comprehend/interpret text by applying critical thinking . • generate questions to understand text.	G09EnU01P01	Analyse Paragraphs and Identify Types of Nouns
	1.1	Write a unified paragraph on a given topic to show • Clear topic sentence using specific words, vivid verbs, modifiers, etc. • Adequate supporting detail (example, illustration, definition, evidence, comparison, contrast, cause and effect, clarification, explanation) to develop the main idea. • Appropriate pronoun-antecedent relationship and transitional devices within a paragraph. • Chronological / sequential / spatial order of arranging detail. • Order of importance (most important to least important and vice versa, general to specific and vice versa).		

	2.1	Demonstrate use of more collective, countable and uncountable, material and abstract nouns.		
	2.1	Apply rules of change of number of nouns learnt earlier.	G09EnU01P02	Agree or Disagree and Make Plurals
	2.1	Demonstrate use of more collective, countable and uncountable, material and abstract nouns.		
	2.1	Recognize and demonstrate use of some nouns that change meaning in the plural form e.g. arm-arms (a part of body), arms (weapons).		
	1.1	Agree and disagree politely at appropriate times		
Lost in Mohenjo-daro	2.1	"Analyse the order of arranging paragraphs: • Chronological or spatial. • General to specific, specific to general. • Most important to least important and vice versa. "	G09EnU02P01	Organise Text, Apologise, and Relate
	1.1	Demonstrate an understanding of ways to show gratitude, apology, anger and impatience		
	2.1	Identify, and demonstrate use of relative pronouns.		
	1.4	Select and use a variety of pre-writing strategies such as brainstorming, mind mapping, outlining etc	G09EnU02P02	Outline Stories and Learn About Verbs
	2.1	Illustrate use of regular and irregular verbs in speech and writing.		
	2.1	Illustrate use of transitive and intransitive verbs.		
Ayubia: The Local Communities	1.2	Use pre-reading strategies to • predict the content of a text from topic / picture, title / headings, key words and visuals etc. by using prior knowledge, asking questions and contextual clues.	G09EnU03P01	Discuss Text and Use Articles
	1.1	Clarify and restate information and ideas		
	2.1	Apply rules for use of a, an and the, wherever applicable in speech and writing.		
	1.2	Use summary skills to write summary/ précis of simple passages / poems.	G09EnU03P02	Summarise Text and Explore Participles
	2.1	Make and use present and past participles.		
	2.1	Make and use present and past participles.		
Gulliver in a New World	2.1	Analyse the order of arranging paragraphs: • Chronological or spatial.	G09EnU04P01	Arrange Paragraphs and Explore Adjectives

		- General to specific, specific to general. - Most important to least important and vice versa.		
	2.1	Classify adjectives into different types. Change and use degrees of adjectives.		
	2.1	Classify adjectives into different types. Change and use degrees of adjectives.		
	2.1	Illustrate use of adverbs learnt earlier.	G09EnU04P02	Everything about Adverbs
	2.1	Identify and use degrees of comparison of adverbs.		
	2.1	Recognize varying positions of adverbs in sentences according to their kinds and importance.		
Samina Baig	1.1	Analyse passages in the text to identify the theme/ general subject, key idea/ central thought (a statement about the general subject), and supporting details.	G09EnU05P01	Analyse Text and Use Commas
	2.2	Recognize and use punctuation including use of commas in phrases, complex clauses and sentences		
	2.1	Illustrate use and functions of all modal verbs.	G09EnU05P02	Use Modal Verbs, Be Grateful, and Learn Prepositions
	1.1	Demonstrate an understanding of ways to show gratitude, apology, anger and impatience		
	2.1	Illustrate the use of prepositions of position, time and movement and direction.		
	2.1	Use prepositional phrases and clauses.		
Hope is the Thing with Feathers	2.1	Read and analyse how a writer/ poet uses language to • appeal to the senses through use of figurative language including similes and metaphors. • affect meaning through use of synonyms with different connotations and denotations. • create imagery.	G09EnU06P01	Analyse Poetry and Revisit Indefinite Tenses
	2.1	Illustrate use of tenses learnt earlier.		
	2.1	Illustrate use of tenses learnt earlier.		
	2.1	Illustrate use of tenses learnt earlier.	G09EnU06P02	Rules of Punctuation and Types of Sentences
	2.1	Illustrate use of punctuation marks learnt earlier.		
	2.3	Analyse sentences for clauses and phrases. Identify and differentiate between main, subordinate and relative clause.		
	2.2	Analyse sentence types and structure, recognize and apply the concept and		

		function of coordination and subordination in extended writing tasks.		
Shahenshah-e-Qawwali	1.1	Analyse paragraphs to identify words, phrases or sentences that support the main idea through • definition • example/ illustration • cause and effect • comparison and contrast • facts, analogies, anecdotes and quotations.	G09EnU07P01	Analyse Paragraphs and Use Pronouns
	2.1	Recognize the rules for using indefinite pronouns.		
	2.1	Illustrate use of infinitives and infinitive phrases.	G09EnU07P02	Understand Infinitives and Write Letters
	1.3	Write and revise formal letters to people in extended social and academic environment for various purposes. Write the address on the envelope clearly and in proper format.		
	1.3	Write and revise applications to people in extended environment using correct format, layout and tone.		
Amelia Earhart	2.2	Skim text to • have general idea of the text. • infer theme/ main idea.	G09EnU08P01	Write Essays and Ask for Instructions
	1.1	Develop focus for own writing by identifying audience and purpose.		
	1.1	Ask and restate and simplify directions and instructions		
		Apply critical thinking to interact with text and use intensive reading strategies (while-reading) to • scan to answer short questions. • make simple inferences using context of the text and prior knowledge. • distinguish between what is clearly stated and what is implied . • deduce meaning of difficult words from context. • use context to infer missing words. • read silently with comprehension and extract main idea and supporting detail. • scan to locate an opinion. • distinguish fact from opinion in letters to editors, texts supporting an opinion, etc. • locate examples to support an opinion e.g. a personal anecdote, quotations, examples and analogy, etc. • recognize arguments and counter arguments. • explore viewpoints/ ideas and issues. • follow instructions in maps or user instruction manuals and forms related to extended social and academic	G09EnU08P02	Connect Thoughts and Express Anger

		environment. • comprehend/interpret text by applying critical thinking . • generate questions to understand text.		
	2.3	Use transitional devices learnt earlier.		
	1.1	Demonstrate an understanding of ways to show gratitude, apology, anger and impatience		
The White Mouse Circus	2.3	Analyse paragraphs to identify words, phrases or sentences that support the main idea through • definition • example/ illustration • cause and effect • comparison and contrast • facts, analogies, anecdotes and quotations.	G09EnU09P01	Use the Right Pronouns and Learn Narration
	2.1	Illustrate use of pronoun-antecedent agreement.		
	2.3	Recognize the rules of and change the narration of statements, requests/ orders and questions.		
	2.2	Analyse story elements: characters, events, setting, plot, theme, tone, point of view.	G09EnU09P02	Change Voice, Use Quotations, and Make Contractions
	2.3	Use active and passive voice appropriately in speech and writing according to the required communicative function.		
	2.2	Recognize and use quotation marks to enclose • both parts of an interrupted quotation. • unusual or peculiar terms to which attention is directed to make the meaning clear.		
Daffodils	2.2	Read and recognize literary techniques such as repetition, personification and alliteration.	G09EnU10P01	Form Adjectives and Greet Informally
	2.1	Form adjectives from nouns and verbs.		
	1.1	Offer and respond to greetings, compliments, invitations, introductions and farewells.		
	2.1	Illustrate use of tenses learnt earlier.	G09EnU10P02	Revise Tenses and Express Gratitude
	2.1	Illustrate use of tenses learnt earlier.		
	1.1	Demonstrate an understanding of ways to show gratitude, apology, anger and impatience		

b. Maths

Unit Name	Contents and Scope	Learning Outcomes /Skills	Skill Name	Package Code	Package Title
MATRICES AND DETERMINANTS	1.1 Introduction to Matrices	Define • a matrix with real entries and relate its rectangular layout (formation) with real life, • rows and columns of a matrix,	Able to define rows and columns of a matrix	G09MU01P01	Types of Matrices and Addition and Subtraction of Matrices
		• the order of a matrix,	Able to define order of matrix		
		• equality of two matrices.	Able to define equality of two matrices		
	1.2 Types of Matrices	Define and identify row matrix, column matrix, rectangular matrix, square matrix, zero/null matrix, identity matrix, scalar matrix, diagonal matrix, transpose of a matrix, symmetric and skew-symmetric matrices.	Row matrix, column matrix, rectangular matrix, square matrix, and zero/null matrix		
			Negative and transpose of a matrix		
			Symmetric matrix and skew-symmetric matrices		
			Identity matrix, diagonal matrix and scalar matrix		
	1.3 Addition and Subtraction of Matrices	i) Know whether the given matrices are conformable for addition/subtraction	Check for conformability for addition/subtraction		

		n. ii) Add and subtract matrices.	and add/subtract matrices		
		iii) Multiply a matrix by a real number.	Multiply a matrix by a real number		
		iv) Verify commutative and associative laws under addition.	Commutative and associative law under addition		
		v) Define additive identity of a matrix.	Additive identity of a matrix		
		vi) Find additive inverse of a matrix.	Find additive inverse of a matrix		
	1.4 Multiplication of Matrices	i) Know whether the given matrices are conformable for multiplication. ii) Multiply two (or three) matrices.	Check for conformability for multiplication and multiply two (or three) matrices	G09MU01P02	Multiplication of Matrices
		iii) Verify associative law under multiplication.	Associative law under multiplication		
		iv) Verify distributive laws.	Distributive laws under multiplication over addition/subtraction		
		v) Show with the help of an example that commutative law under multiplication does not hold in general (i.e. $AB \neq BA$).	Commutative law under multiplication does not hold in general		
		vi) Define multiplicative identity of a matrix.	Multiplicative identity of a matrix		
		vii) Verify the result $(AB)^t = B^t A^t$.	Verify the result $(AB)^t = B^t A^t$		
	1.5 Multiplicative Inverse of a Matrix	i) Define the determinant of a square matrix.	Find determinant of a 2-by-2 matrix	G09MU01P03	Determinant, Adjoint, and Multiplicative

		ii) Evaluate determinant of a matrix.	Evaluate determinant of a matrix		Inverse of Matrices
		iii) Define singular and non-singular matrices.	Define singular and non-singular matrices		
		iv) Define adjoint of a matrix.	Define and find adjoint of a matrix		
		v) Find multiplicative inverse of a non-singular matrix A and verify that $AA^{-1} = I = A^{-1}A$ where I is the identity matrix.	Find multiplicative inverse of a non-singular matrix A		
		vi) Use adjoint method to calculate inverse of a nonsingular matrix.	Find inverse of a matrix using adjoint		
		vii) Verify the result $(AB)^{-1} = B^{-1}A^{-1}$.	Verify the result $(AB)^{-1} = B^{-1}A^{-1}$		
	1.6 Solution of Simultaneous Linear Equations	Solve a system of two linear equations and related real life problems in two unknowns using - Matrix inversion method, - Cramer's rule.	Solve a system of two linear equations using "Matrix inversion method"		
			Solve a system of two linear equations using "Cramer's rule"		
REAL AND COMPLEX NUMBERS	2.1 Real Numbers	i) Recall the set of real numbers as a union of sets of rational and irrational numbers.	Understand real numbers as a union of sets of rational and irrational numbers	G09MU02P01	Real Numbers and Their Properties Part 1
		ii) Depict real numbers on the number line.	Depict real numbers on		

			the number line		
		iii) Demonstrate a number with terminating and non-terminating recurring decimals on the number line.	Depict a number with terminating/non-terminating recurring decimals on the number line		
		iv) Give decimal representation of rational and irrational numbers.	Give decimal representation of rational and irrational numbers		
	2.2 Properties of Real Numbers	Know the properties of real numbers.	Closure / commutative / associative property, additive identity/inverse under + / -		
			Closure / commutative / associative / distributive property, multiplicative identity/inverse under x		
			Reflexive / symmetric / transitive / additive / multiplicative and cancellation property for + / x	G09MU02P02	Real Numbers and Their Properties Part 2
			Trichotomy / transitive / additive / multiplicative and multiplicative inverse property for inequality		

	2.3 Radicals and Radicands	i) Explain the concept of radicals and radicands. ii) Differentiate between radical form and exponential form of an expression.	Explain the concept of radicals and radicands	G09MU02P03	Radicals, Radicands, and Laws of Exponents
		iii) Transform an expression given in radical form to an exponential form and vice versa.	Transform an expression between radical form and exponential form		
	2.4 Laws of Exponents/Indices	i) Recall base, exponent and value.	Base, exponent, and value		
		ii) Apply the laws of exponents to simplify expressions with real exponents.	Apply the laws of exponents to simplify expressions with real exponents		
	2.5 Complex Numbers	i) Define complex number z represented by an expression of the form $z = a + ib$, where a and b are real numbers and $i = -1$.	Define complex number z represented by an expression of the form $z = a + ib$	G09MU02P04	Complex Numbers
		ii) Recognize a as real part and b as imaginary part of $z = a + ib$.	Recognize a as real part and b as imaginary part of z		
		iii) Define conjugate of a complex number.	Define conjugate of a complex number		
		iv) Know the condition for equality of complex numbers.	Know the condition for equality of complex numbers		
	2.6 Basic Operations on Complex Numbers	Carryout basic operations (i.e. addition, subtraction, multiplication and	Addition/subtraction on complex numbers		

		division) on complex numbers.			
			Multiplication on complex numbers		
			Division on complex numbers		
LOGARITHMS	3.1 Scientific Notation	Express a number in standard form of scientific notation and vice versa.	Express a number in standard form of scientific notation and vice versa	G09MU03P01	Scientific Notation and Logarithms
	3.2 Logarithm	i) Define logarithm of a number to the base a as the power to which a must be raised to give the number (i.e. $ax = y \Leftrightarrow \log_y x, a = a > 0, y > 0$ and $a \neq 1$).	Define logarithm of a number y to the base a as the power to which a must be raised to give the number		
		ii) Define a common logarithm, characteristic and mantissa of log of a number.	Define characteristic and mantissa of common log of a number		
		iii) Use tables to find the log of a number.	Use tables to find the log of a number		
		iv) Give concept of antilog and use tables to find the antilog of a number.	Give concepts of anti-log and use tables to find antilog of a number		
	3.3 Common and Natural Logarithm	Differentiate between common and natural logarithm.	Differentiate between common and natural logarithm	G09MU03P03	Common and Natural Logarithm, Antilog and Laws of Logarithms
	3.4 Laws of Logarithm	Prove the law of logarithm: $\log_a(mn) = \log_a m + \log_a n$	Prove the law of logarithm: $\log_a(mn) = \log_a m + \log_a n$		
		Prove the law of logarithm: $\log_a(m/n) = \log_a m - \log_a n$	Prove the law of logarithm: $\log_a(m/n) = \log_a m - \log_a n$		

		Prove the law of logarithm: $\log_{am}n = n \log_a m$	Prove the law of logarithm: $\log_{am}n = n \log_a m$	G09MU03P04	Laws of Logarithms and Their Applications
		Prove the law of logarithm: $\log_a m \log_m n = \log_a n$	Prove the law of logarithm: $\log_a m \log_m n = \log_a n$		
	3.5 Application of Logarithm	Apply laws of logarithm to convert lengthy processes of multiplication, division and exponentiation into easier processes of addition and subtraction etc.	Apply laws of logarithm to simplify lengthy processes		
ALGEBRAIC EXPRESSIONS AND ALGEBRAIC FORMULAS	4.1 Algebraic expressions	i) Know that a rational expression behaves like a rational number.	Know that a rational expression behaves like a rational number	G09MU04P01	Rational Expressions
		ii) Define a rational expression as the quotient $p(x)/q(x)$ of two polynomials $p(x)$ and $q(x)$ where $q(x)$ is not the zero polynomial.	Define a rational expression as the quotient $p(x)/q(x)$ of two polynomials $p(x)$ and $q(x)$		
		iii) Examine whether a given algebraic expression is a • polynomial or not, • rational expression or not.	Examine whether a given algebraic expression is a rational expression		
			Check whether a given algebraic expression is a polynomial		
		iv) Define $p(x)/q(x)$ as a rational expression in its lowest terms if $p(x)$ and $q(x)$ are	Define $p(x)/q(x)$ as a rational expression in its lowest		

		polynomials with integral coefficients and having no common factor.	terms if $p(x)$ and $q(x)$ are polynomials with integral coefficients and having no common factor.		
		v) Examine whether a given rational algebraic expression is in lowest form or not. vi) Reduce a given rational expression to its lowest terms.	Reduce a given algebraic expression to its lowest form		
		vii) Find the sum, difference and product of rational expressions.	Find the sum, difference and product of rational expressions	G09MU04P02	Solving Rational Expressions
		viii) Divide a rational expression with another and express the result in its lowest terms.	Divide a rational expression with another and express the result in its lowest terms		
		ix) Find value of algebraic expression at some particular real number.	Evaluate an algebraic expression when the values of variables involved are given		
	4.2 Algebraic Formulae	Know the formulas: $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$ and $(a+b)^2 - (a-b)^2 = 4ab$	$(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$ and $(a+b)^2 - (a-b)^2 = 4ab$	G09MU04P03	Algebraic Formulae
		Find value of $a^2 + b^2$ and of ab when values of $a + b$ and $a - b$ are known	Find value of $a^2 + b^2$ and of ab when values of $a + b$ and $a - b$ are known		
		Know the formula: $(a+b+c)^2 =$	$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$		

		$a^2+b^2+c^2+2ab+2bc+2ca$			
		Find value of $a^2 + b^2 + c^2$ when $a + b + c$ and $ab + bc + ca$ are given and vice versa	Find value of $a^2 + b^2 + c^2$ when $a + b + c$ and $ab + bc + ca$ are given and vice versa		
		Find $a + b + c$ when $a^2+b^2+c^2$ and $ab+bc+ca$ are given	Find $a + b + c$ when $a^2+b^2+c^2$ and $ab+bc+ca$ are given		
		Find $ab + bc + ca$ when $a^2+b^2+c^2$ and $a+b+c$ are given	Find $ab + bc + ca$ when $a^2+b^2+c^2$ and $a+b+c$ are given		
		Know the formula: $(a \pm b)^3 = a^3 \pm 3ab(a \pm b) \pm b^3$	Formula: $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$	G09MU04P04	Algebraic Formulae Involving Cubes
			Formula: $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$		
		Find $a^3 + b^3$ given $(a+b)$ or $(a-b)$ and ab	Find $a^3 + b^3$ given $(a+b)$ or $(a-b)$ and ab		
		Find $x^3 + 1/x^3$ when the values of $x + 1/x$ are given	Find $x^3 + 1/x^3$ given $x + 1/x$		
		Know the formula: $a^3 \pm b^3 = (a \pm b)(a^2 + ab + b^2)$	$a^3 \pm b^3 = (a \pm b)(a^2 + ab + b^2)$	G09MU04P05	Cubic Formulae
		Find the product of $(x+1/x)(x^2 + 1/x^2 - 1)$; Find the product of $(x - 1/x)(x^2 + 1/x^2 + 1)$	Find $(x + 1/x)(x^2 + 1/x^2 - 1)$		
		Find the continued product of $(x+y)(x-y)(x^2+xy+y^2)(x^2-xy+y^2)$	Find $(x+y)(x-y)(x^2+xy+y^2)(x^2-xy+y^2)$		
4.3 Surds and their Application	i) Recognize the surds and their application.	Introduction to surds and types of surds		G09MU04P06	Surds and Their Applications

		ii) Explain the surds of second order. Use basic operations on surds of second order to rationalize the denominators and evaluate it.	Addition and subtraction of surds		
			Multiplication and division of surds		
	4.4 Rationalization	Explain rationalization (with precise meaning) of real numbers of the types $1/a+bx$, $1/vx + vy$ and their combinations where x and y are natural numbers and a and b are integers	Rationalize the denominators and evaluate surds of second order		
			Rationalize $1/(a+b\sqrt{x})$, $1/(\sqrt{x} + \sqrt{y})$ and their combinations		
FACTORIZATION	5.1 Factorization	Recall factorization of expressions of the following types. • $ka + kb + kc$,	Factorise $ka + kb + kc$	G09MU05P01	Factorising Algebraic Expressions Part 1
		Factorize $ac + ad + bc + bd$	Factorize $ac + ad + bc + bd$		
		Factorize $a^2 + 2ab + b^2$	Factorize $a^2 + 2ab + b^2$		
			Factorize an algebraic expression by using algebraic identities	G09MU05P02	Factorising Algebraic Expressions Part 2
		Factorize $a^2 - b^2$	Factorize $a^2 - b^2$		
		Factorize $a^2 + 2ab + b^2 - c^2$	Factorize $a^2 + 2ab + b^2 - c^2$		

		Factorize $a^4 + a^2b^2 + b^4$ or $a^4 + 4b^4$	Factorize $a^4 + a^2b^2 + b^4$ or $a^4 + 4b^4$	G09MU05P03	Factorising Algebraic Expressions Part 3
		Factorize x^2+px+q	Factorize x^2+px+q		
		Factorize $ax^2 + bx + c$	Factorize $ax^2 + bx + c$		
		Factorize $(ax^2+bx+c)(ax^2+bx+d)+k$ or $(x+a)(x+b)(x+c)(x+d)+k$ or $(x+a)(x+b)(x+c)(x+d)+kx^2$	Factorize $(ax^2+bx+c)(ax^2+bx+d)+k$ or $(x+a)(x+b)(x+c)(x+d)+k$ or $(x+a)(x+b)(x+c)(x+d)+kx^2$		
			Formula: $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$	G09MU05P04	Factorise Algebraic Expressions Involving Cubes
			Formula: $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$		
		Factorize $a^3 + 3a^2b + 3ab^2 + b^3$	Factorize $a^3 + 3a^2b + 3ab^2 + b^3$		
		Factorize $a^3 - 3a^2b + 3ab^2 - b^3$	Factorize $a^3 - 3a^2b + 3ab^2 - b^3$		
		Factorize $a^3 + b^3$	Factorize $a^3 + b^3$		
	5.2 Remainder Theorem and Factor Theorem	i) State and prove Remainder theorem and explain through examples.	Remainder Theorem	G09MU05P05	Remainder Theorem and Factor Theorem
		ii) Find remainder (without dividing) when a polynomial is divided by a linear polynomial.	Find remainder (without dividing) when a polynomial is divided by a linear polynomial		
		iii) Define zeros of a polynomial.	Define zeros of a polynomial		
		iv) State and prove factor theorem.	Factor Theorem		

	5.3 Factorization of a Cubic Polynomial	Use factor theorem to factorize a cubic polynomial.	Use factor theorem to factorize a cubic polynomial		
ALGEBRAIC MANIPULATION	6.1 Highest Common Factor and Least Common Multiple	i) Find highest common factor and least common multiple of algebraic expressions.	Find HCF by factorization	G09MU06P01	HCF and LCM
			Find LCM by factorization		
		ii) Use factor or division method to determine highest common factor and least common multiple.	Find HCF by division		
		iii) Know the relationship between HCF and LCM.	Know the relationship between HCF and LCM	G09MU06P02	Application of HCF and LCM
		iv) Solve real life problems related to HCF and LCM.	Solve real life problems related to HCF and LCM		
	6.2 Basic Operations on Algebraic Fractions	Use highest common factor and least common multiple to reduce fractional expressions involving $+, -, \times, \div$.	Use HCF and LCM to reduce fractional expressions involving $+, -, \times, /$		
LINEAR EQUATIONS AND INEQUALITIES	7.1 Linear Equations		Solve linear equation with one variable	G09MU07P01	Linear Equations
		ii) Solve linear equation with rational coefficients.			
		iii) Reduce equations, involving radicals, to simple linear form and find their solutions.	Solve equations involving radicals but reducible to linear form		
	7.2 Equation involving Absolute Value	i) Define absolute value. ii) Solve the equation, involving	Solve linear equations involving absolute value		

		absolute value, in one variable.			
	7.3 Linear Inequalities	i) Define inequalities ($>$, $<$) and ($\geq$, $\leq$).	Define inequalities ($>$, $<$) and ($\leq$, $\geq$)	G09MU07P02	Linear Inequalities
		ii) Recognize properties of inequalities (i.e. trichotomy, transitive, additive and multiplicative).	Recognize properties of inequalities. (trichotomy, transitive, additive and multiplicative)		
	7.4 Solving Linear Inequalities	Solve linear inequalities with rational coefficients.	Solve linear inequalities with rational coefficients		
LINEAR GRAPHS AND THEIR APPLICATION	14.1 Cartesian Plane and Linear Graphs	i) Identify pair of real numbers as an ordered pair. ii) Recognize an ordered pair through different examples; for instance an ordered pair (2,3) to represent a seat, located in an examination hall, at the 2nd row and 3rd column.	Recognize an ordered pair through different examples	G09MU08P01	Cartesian Plane and Linear Graphs Part 1
		iii) Describe rectangular or Cartesian plane consisting of two number lines intersecting at right angles at the point O. iv) Identify origin (O) and coordinate axes (horizontal and vertical axes or x-axis and y-axis) in the rectangular plane.	Recognize Cartesian plane, identification of origin and coordinate axes		
		v) Locate an ordered pair (a, b) as a point in the rectangular plane and recognize: • a as the x-coordinate	Recognize x-coordinate (or abscissa) and y-coordinate (or ordinate)		

		(or abscissa), • b as the y-coordinate (or ordinate).			
		vi) Draw different geometrical shapes (e.g., line segment, triangle and rectangle etc) by joining a set of given points.	Draw different geometrical shapes in Cartesian plane (line, triangle, rectangle)		
		vii) Construct a table for pairs of values satisfying a linear equation in two variables.	Construct a table for pairs of values satisfying a linear equation in two variables		
		viii) Plot the pairs of points to obtain the graph of a given expression.	Plot the pairs of points to obtain the graph of a given expression		
		ix) Choose an appropriate scale to draw a graph.	Choose an appropriate scale to draw a graph	G09MU08P02	Cartesian Plane and Linear Graphs Part 2
		x) Draw the graph of • an equation of the form $y = c$.	Use appropriate scale, draw a graph of equation of $y = c$		
		• an equation of the form $x = a$.	Use appropriate scale, draw a graph of equation of $x = a$		
		• an equation of the form $y = mx + c$.	Use appropriate scale, draw a graph of equation of $y = mx + c$		
		• an equation of the form $y = mx$.	Draw the graph of an equation of the form $y = mx$		

		xi) Draw a graph from a given table of (discrete) values.	Use appropriate scale, draw a graph from a given table of (discrete) values	G09MU08P03	Conversion Graphs
	14.2 Conversion Graphs	i) Interpret conversion graph as a linear graph relating to two quantities which are in direct proportion.	Interpret conversion graph as a linear graph relating to two proportional quantities		
		ii) Read a given graph to know one quantity corresponding to another.	Read a given graph to know one quantity corresponding to another		
		iii) Read the graph for conversions of the form: • miles and kilometers, • acres and hectares, • degrees Celsius and degrees Fahrenheit, • Pakistani currency and another currency, etc.	Read the graph for conversions		
	14.3 Graphic Solution of Equations in two Variables	Solve simultaneous linear equations in two variables using graphical method.	Solve real life problems involving two simultaneous linear equations in two variables		
			Solve simultaneous linear equation in two variable using graphical method		
INTRODUCTION TO COORDINATE GEOMETRY	15.1 Distance Formula	i) Define coordinate geometry.	Define coordinate geometry	G09MU09P01	Distance Formula and Collinear Points
		ii) Derive distance formula to calculate distance between	Calculate distance between two		

		two points given in Cartesian plane. iii) Use distance formula to find distance between two given points.	points given in Cartesian plane		
	15.2 Collinear Points	i) Define collinear points. Distinguish between collinear and non-collinear points.	Define collinear points		
			Distinguish between collinear and non-collinear points		
		ii) Use distance formula to show that given three (or more) points are collinear.	Show that given three points are collinear in a plane		
		iii) Use distance formula to show that the given three non-collinear points form: • an equilateral triangle,	Show that the given three non-collinear points form an equilateral triangle		
		• an isosceles triangle,	Show that the given three non-collinear points form an isosceles triangle	G09MU09P02	Use Distance Formula to Form Different Shapes of a Triangle
		• a right angled triangle,	Show that the given three non-collinear points form a right angled triangle		
		• a scalene triangle.	Show that the given three non-collinear points form a scalene triangle		
		iv) Use distance formula to show that given four noncollinear points form: • a square,	Show that given four non-collinear points form a square	G09MU09P03	Using Distance Formula to Form Different Quadrilaterals

		• a rectangle,	Show that given four non-collinear points form a rectangle		
		• a parallelogram.	Show that given four non-collinear points form a parallelogram		
	15.3 Mid-point Formula	i) Recognize the formula to find the midpoint of the line joining two given points.	Find the midpoint of the line joining two given points	G09MU09P04	Distance and Mid-Point Formula
		ii) Apply distance and mid point formulae to solve/verify different standard results related to geometry.	Apply distance and mid-point formulae in geometry		
CONGRUENT TRIANGLES	17.1 Congruent Triangles	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) In any correspondence of two triangles, if one side and any two angles of one triangle are congruent to the corresponding side and angles of the other, the two triangles are congruent.	Prove congruency of two triangles based on ASA	G09MU10P01	Congruent Triangles Part 1
		ii) If two angles of a triangle are congruent then the sides opposite to them are also congruent.	Prove that in a triangle, sides opposite to congruent angles are congruent		
		iii) In a correspondence of two triangles, if three sides of one triangle are	Prove congruency of two triangles based on SSS		

		congruent to the corresponding three sides of the other, the two triangles are congruent.			
		iv) If in the correspondence of two right-angled triangles, the hypotenuse and one side of one are congruent to the hypotenuse and the corresponding side of the other, then the triangles are congruent.	Prove congruency of two triangles based on HL	G09MU10P02	Congruent Triangles Part 2
PARALLELOGRAMS AND TRIANGLES	18.1 Parallelograms and Triangles	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) In a parallelogram: - the opposite sides are congruent, - the opposite angles are congruent, - the diagonals bisect each other.	Prove properties of parallelogram	G09MU11P01	Theorems Related to Parallelograms and Triangles
		ii) If two opposite sides of a quadrilateral are congruent and parallel, it is a parallelogram.	Prove a quadrilateral is a parallelogram		
		iii) The line segment, joining the midpoints of two sides of a triangle, is parallel to the third side and is equal to one half of its length.	Prove triangle mid-point properties		
		iv) The medians of a triangle are concurrent and their point of concurrency is the	Prove triangle median properties	G09MU11P02	Theorems Related to Triangles and Parallel Lines

		point of trisection of each median.			
		v) If three or more parallel lines make congruent intercepts on a transversal they also intercept congruent segments on any other line that cuts them.	Prove parallel line properties		
LINE BISECTORS AND ANGLE BISECTORS	19.1 Line Bisectors and Angle Bisectors	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) Any point on the right bisector of a line segment is equidistant from its end points.	Properties of right bisector of a line segment	G09MU12P01	Line and Angle Bisectors Part 1
		ii) Any point equidistant from the points of a line segment is on the right bisector of it.	Proof of any point that lies on the right bisector of a line segment		
		iii) The right bisectors of the sides of a triangle are concurrent.	Prove that the right bisectors of the sides of a triangle are concurrent		
		iv) Any point on the bisector of an angle is equidistant from its arms.	Prove that any point on the bisector of an angle is equidistant from its arms		
		v) Any point inside an angle, equidistant from its arms, is on the bisector of it.	Prove that any point inside an angle, equidistant from its arms, is on the bisector of it	G09MU12P02	Line and Angle Bisectors Part 2
		vi) The bisectors of the angles of a triangle are concurrent.	Prove that the bisectors of the angles of a		

			triangle are concurrent		
SIDES AND ANGLES OF A TRIANGLE		Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) If two sides of a triangle are unequal in length, the longer side has an angle of greater measure opposite to it.	Prove that in a triangle, the longer side has an angle of greater measure opposite to it	G09MU13P01	Sides and Angles of a Triangle
	20.1 Sides and Angles of a Triangle				
		ii) If two angles of a triangle are unequal in measure, the side opposite to the greater angle is longer than the side opposite to the smaller angle.	Prove that in a triangle, the side opposite to the greater angle is longer than the other sides		
		iii) The sum of the lengths of any two sides of a triangle is greater than the length of the third side.	Prove that the sum of the lengths of any two sides in a triangle is greater than the third		
		iv) From a point, out-side a line, the perpendicular is the shortest distance from the point to the line.	Prove that the perpendicular is the shortest distance from any point to the line		
RATIO AND PROPORTION		Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) A line parallel to one side of a triangle, intersecting the other two sides, divides them proportionally.	Prove that a line parallel to one side of a triangle, cut the other two sides proportionally	G09MU14P01	Theorems on Ratios and Proportions
	21.1 Ratio and Proportion				
		ii) If a line segment intersects the two	Prove a line segment cuts		

		sides of a triangle in the same ratio then it is parallel to the third side.	two sides of a triangle in the same ratio is parallel to the third side		
		iii) The internal bisector of an angle of a triangle divides the side opposite to it in the ratio of the lengths of the sides containing the angle.	Prove that triangle angle bisector divides opposite side proportionally to arms of the angle		
		iv) If two triangles are similar, the measures of their corresponding sides are proportional.	Prove that if two triangles are similar, the length of their corresponding sides are proportional		
PYTHAGORAS' THEOREM	22.1 Pythagoras' Theorem	Prove the following theorems along with corollaries and apply them to solve appropriate problems. i) In a right-angled triangle, the square of the length of hypotenuse is equal to the sum of the squares of the lengths of the other two sides. (Pythagoras' theorem).	Prove Pythagoras theorem for right angled triangle	G09MU15P01	Pythagoras' Theorem
		ii) If the square of one side of a triangle is equal to the sum of the squares of the other two sides then the triangle is a right angled triangle (converse to Pythagoras' theorem).	Prove a triangle is right angled triangle based on reverse of Pythagoras theorem		
THEOREMS RELATED WITH AREA	23.1 Theorems Related with Area	Prove the following theorems along with corollaries and	Parallelograms and their Properties	G09MU16P01	Theorems Related With Area

		apply them to solve appropriate problems. i) Parallelograms on the same base and lying between the same parallel lines (or of the same altitude) are equal in area.			
			Prove that parallelograms on the same base between parallel lines are equal in area		
		ii) Parallelograms on equal bases and having the same altitude are equal in area.	Prove that parallelograms on equal bases and same altitude are equal in area		
		iii) Triangles on the same base and of the same altitude are equal in area.	Prove that triangles on the same base and of the same altitude are equal in area		
		iv) Triangles on equal bases and of the same altitude are equal in area.	Prove that triangles on equal bases and of the same altitude are equal in area		
PRACTICAL GEOMETRY – TRIANGLES	29.1 Construction of Triangle	i) Construct a triangle having given: • two sides and the included angle,	Construct a triangle given two sides and the included angle	G09MU17P01	Constructing Triangles Part 1
		• one side and two of the angles,	Construct a triangle given one side and two of the angles		
		• two of its sides and the angle opposite to one of	Construct a triangle given two sides and		

		them (with all the three possibilities).	the angle opposite to one of them		
		ii) Draw: • angle bisectors,	Draw angle bisectors of a given triangle and verify their concurrency		
		• altitudes,	Draw altitudes of a given triangle and verify their concurrency		
		• perpendicular bisectors,	Draw perpendicular bisectors of a given triangle and verify their concurrency	G09MU17P02	Constructing Triangles Part 2
		• medians, of a given triangle and verify their concurrency.	Draw medians of a given triangle and verify their concurrency		
	29.2 Figures with Equal Areas	i) Construct a triangle equal in area to a given quadrilateral.	Construct a triangle equal in area to a given quadrilateral		
		ii) Construct a rectangle equal in area to a given triangle.	Construct a rectangle equal in area to a given triangle		
		iii) Construct a square equal in area to a given rectangle.	Construct a square equal in area to a given rectangle		
		iv) Construct a triangle of equivalent area on a base of given length.	Construct a triangle of equal area on the base of given length		

c. Biology

Unit	SLO	Package Code	Package Name
INTRODUCTION TO BIOLOGY	Define biology its major divisions i.e. botany, zoology and microbiology.	G09BU01P01	Biology and Its Branches
	Define the branches of biology i.e. morphology, anatomy, physiology, embryology, taxonomy, cell biology, histology, paleontology, environmental biology, biotechnology, socio-biology, parasitology, immunology, entomology, genetics, pharmacology.		
	Link the study of biology with that of physics, chemistry, mathematics, geography and economics.		
	Explain how the study of biology can lead to Medicine / Surgery, Fisheries, Agriculture, Animal husbandry, Biotechnology, Horticulture, Farming, Forestry.		
	Identify that living organisms are divided in five groups i.e. prokaryotes, protists, fungi, plants and animals.		
	Describe bioelements as the most the most basic level of biological organization.	G09BU01P02	Biological Levels of Organisation
	Define biomolecules and distinguish them as micromolecules and macromolecules.		
	Describe the level of organization of life (organelles, cells, tissues, organs and organ systems and individuals).		
	Explain division of labor among cells and tissues in a multicellular organism.		
	Compare cellular organization in organisms i.e. unicellular organization (Amoeba), colonial organization (Volvox) and multicellular organization (mustard and frog). (Only brief comparison referring to cellular organization is required. Details of organs and organs-systems of frog and mustard should be avoided)	G09BU01P03	Unicellular and Multicellular Organisms
BIODIVERSITY	Define biodiversity.	G09BU03P01	Biodiversity, Aims and Principles of Classification
	Describe the major variety of life on the planet earth.		

	Relate the importance of biodiversity with natural ecosystem through examples		
	Explain the aims and principles of classification, keeping in view its historical background.		
	Explain the bases for establishing 5 kingdoms.		
	Describe the aims and principles binomial nomenclature keeping in view the historical background.		
	Describe using local examples, the importance of Binomial nomenclature.		
	Describe the bases of classification of living organisms.	G09BU03P03	Systems of Classification
	Compare Two-kingdom and Five-kingdom classification systems.		
	Rationalize that Five-kingdom classification system better explains diversity of living organisms.		
	Describe the diagnostic characteristics of the five kingdoms.		
	Describe the acellular structure of virus and justify why virus are excluded from the FiveKingdom classification system.		
	Rationalize that there are sub-cellular particles, such as viruses and prions, which have some characteristics of living things.		
	Define the concept of conservation.		
	Explain the impact of human beings on biodiversity.		
	Identify causes of deforestation and its effects on biodiversity.		
	Enumerate the reasons for conservation of biodiversity.		
CELLS AND TISSUES	Identify the structure and describe, in general terms, the functions of the components of plant and animal cell.	G09BU04P02	Cellular Structures and Their Functions
	Assess the capabilities of animal and plant cell types, owing to the presence or absence of chloroplasts and cell wall.		
	Describe the role of the cell membrane in maintaining equilibrium while exchanging matter.		
	Justify how the cells of the leaf system have a variety of specialized structures and functions.		
	State the relationship between cell function and cell structure (for absorption - root hair cells; conduction and support - xylem vessels; transport of oxygen - red blood cells).		
ENZYMES	Define metabolism and differentiate between catabolism and anabolism.	G09BU06P01	Enzymes

	Describe enzymes as the proteins that speed up biochemical reactions.		
	Categorize enzymes as intra and extracellular.		
	Comprehend that enzymes increase rate of reaction.		
	State that small quantity of enzyme is effective for large amount of substrate.		
	Infer that enzymes are specific for specific substrates.		
	Categories that proteases will act on proteins only and lipases will act on lipids or fats only		
	Describe the concept of energy of activation and how it is lowered by enzyme.		
	Describe, through equation, that enzyme substrate complex is formed and release of enzyme takes place after completing the reaction.		
	Describe the action of enzyme through Lock-n-Key Model.		
	Describe the specificity of different enzymes for different substrates.		
	Relate that specificity of enzyme is due to its shape.		
	State that some enzymes require co-factor for their functioning.		
	Explain the effect of pH, temperature and concentration of substrate on the activity of an enzyme.		
BIOENERGETICS	Describe, in general terms, the intake of carbon dioxide and water by plants.	G09BU07P02	Photosynthesis
	State that photosynthesis is the fundamental process by which plants manufacture carbohydrates from raw materials.		
	State the equation (in words or symbols) for photosynthesis.		
	Reason out that all forms of life are completely dependent on photosynthesis.		
	Describe that chlorophyll traps light energy and converts it into chemical energy for the formation of carbohydrates and their subsequent storage.		
	Describe, in general terms, the intake of carbon dioxide and water by plants.	G09BU07P03	Respiration
	Describe aerobic respiration by means of word and symbol equation.		
	Describe anaerobic respiration by means of word and symbol equation.		
	Describe the importance of Anaerobic Respiration.		

	Compare aerobic and anaerobic respiration with reference to the amount of energy released.		
	List ways in which respiratory energy is used in the body.		
NUTRITION	Define mineral nutrition in plants.	G09BU08P01	Macro and Micro Nutrients
	Categorize minerals nutrients into macronutrients and micronutrients.		
	State that nitrogen is important in protein synthesis and magnesium for chlorophyll formation.		
	State the effect of lack of nitrate and magnesium ions on plant growth.		
	Describe the importance of fertilizers (manure and chemical) in agriculture.		
	Discuss environmental hazards related to chemical fertilizers' use.		
	Distinguish among carbohydrates, proteins and fats in terms of their sources, energy values and metabolic functions.	G09BU08P02	Balanced Diet and Digestion
	Describe the concept and need for a balanced diet.		
	Explain the components of a balanced diet with relation to age, sex and activity.		
	Explain why diet, especially energy intake, should be related to age, sex and activity of an individual.		
	Describe the needs of ingestion, digestion, absorption, assimilation and egestion.	G09BU08P04	Structure and Function of Alimentary Canal
	Identify and describe the structures of the main regions of the alimentary canal and the associated organs.		
	Describe the main functions of these parts in relation to ingestion, digestion, absorption, assimilation and egestion of food.		

	Describe swallowing and peristalsis.		
	Sort out the action of enzymes in specific regions of alimentary canal, with respect to their substrates and products.		
	Describe the structure of a villus, including the roles of capillaries and lacteals.		
	Describe the significance of villi in increasing the internal surface area.		
	State the role of the liver in the metabolism of glucose and amino acids, and in the formation of urea.		
TRANSPORT	Explain the internal structure of root and root hair.	G09BU09P01	Transport in Plants
	Describe how roots take up water and mineral salts by active and passive absorption.		
	Define transpiration and relate this process with cell surface.		
	Relate transpiration with stomatal opening and closing.		
	Describe temperature, wind and humidity as the factors affecting the rate of transpiration.		
	Describe the significance of transpiration.		
	Describe transpiration as a necessary evil.		
	Explain the movement of water in terms of transpirational pull.	G09BU09P02	Food Translocation (Plants) and Blood Circulation (Animals)
	Explain the mechanism of food translocation by the theory of Pressure Flow Mechanism.		
	List the functions of the components of blood.		
	Compare the structure and function of an artery, a vein and a capillary.		
	Describe the transfer of materials between capillaries and tissue fluid.	G09BU09P04	Human Circulatory System
	Describe the major pathway of blood through circulatory system.		
	Describe the low-pressure circulation to the lungs and a high-pressure circulation to the body tissues and relate these differences to the different functions of the two circuits.		
	Describe the external and internal structure of human heart.		
	Describe the circulation of blood through atria and ventricles of the heart, explaining the role of the bicuspid, tricuspid and semilunar valves.		
	Explain how the heart is structurally adapted to its functions.		

	Define the terms heartbeat, heart rate and pulse rate.		
	State the causes, treatments and prevention of Myocardial infarction.		
SOLVING BIOLOGICAL PROBLEM	Describe the steps involved in biological method i.e. recognition of a biological problem, observation and identification, building up hypotheses, drawing deductions, devising experiments and inferring results (malaria as an example).	G09BU02P01 (partial)	Biological Method
	Describe the use of ratio and proportion in solving biological problems.		
	Justify mathematics as an integral part of the scientific process.		
	Explain the importance of data analysis for confirming, modifying, or rejecting a hypothesis.		
CELLS AND TISSUES	Explain the concepts of light microscopy and electron microscopy.	G09BU04P01 (partial)	Electron and Compound Microscopes
	Determine ways in which various types of cells contribute to the healthy functioning of the human body (e.g., describe the roles of individual cells in nerves, muscle, blood, skin and bone).	G09BU04P03 (partial)	Prokaryotic and Eukaryotic Cells
	Describe the major animal tissues (epithelial, connective, muscular and nervous) in terms of their cell specificities, locations and functions.		
	Relate the function of a particular tissue to its structure and ultimately to the structure and function of the cells constituting it.		
	Describe the differences in the structure and function of Prokaryotic and Eukaryotic Cells.		
	Assess the capabilities of Prokaryotic and Eukaryotic Cells, owing to the presence or absence of nucleus and mitochondria.		
NUTRITION	Rationalize the unequal distribution of food, drought and flooding, and increasing population as the factors that contribute to famine.	G09BU08P05 (partial)	Famine Determinants
CELLS AND TISSUES	Describe the phenomena of diffusion, facilitated diffusion, osmosis, filtration, active transport, endocytosis and exocytosis.	G09BU04P04	Diffusion, Osmosis, Active Transport and Turgidity
	Compare passive transport of matter by diffusion and osmosis with active transport (e.g. Diffusion of glucose from intestine to villus epithelium and active transport of Sodium ions from nerve cell to outside.)		
	Define turgor and describe its importance.		
	Describe the phenomena of plasmolysis and explain its relationship with osmosis.		

CELL CYCLE	Define Cell Cycle and describe its main phases i.e. Interphase and Division.	G09BU05P01	Cell Cycle
	Enlist the events through which Mitotic Apparatus is formed in prophase in animal and plant cells.		
	Describe the formation of metaphase plate and the division of centromere, during metaphase.		
	State the separation of chromatids during anaphase.		
	Describe the reformation of nuclei during telophase.		
	Describe the physical division of cytoplasm during cytokinesis in animal and plant cells.		
	Compare the details of events during mitosis in animal and plant cells.		
	Describe the significance of mitosis as giving rise to genetically identical cells and state the role of mitosis in growth, repair of damaged tissues, replacement of worn out cells and asexual reproduction.		
	Describe the events of Prophase-I.		
	Describe the events taking place in Metaphase-I.		
	Explain what happens during Anaphase-I.		
	Describe the events of Telophase-I.		
	Explain the events occurring during the Second Meiotic Division.		
	Describe the significance of meiosis as leading to the formation of haploid cells, that may function directly as gametes as in animals or may divide by mitosis as in plants, fungi and many protists.		
	Describe the significance of meiosis with reference to the recombination of genes that leads to variations.		
BIOENERGETICS	Describe the two types of Cell Division in Eukaryotic cells i.e. Mitosis and Meiosis.	G09BU07P04	Photosynthesis and Respiration
	Compare the Second Meiotic division with mitosis.		
	Contrast mitosis and meiosis, emphasizing the events that lead to different outcomes.		
	Explain the concept of limiting factors in photosynthesis.		

	State the effect of varying light intensity, carbon dioxide concentration and temperature on the rate of photosynthesis.		
	Compare respiration and photosynthesis.		
NUTRITION	State the signs and symptoms, causes, treatments and preventions of the disorders of gut i.e. diarrhea, constipation, and ulcer.	G09BU08P06	Disorders of Gut

d. Chemistry

Unit	SLO	Package Code	Package Name
Fundamentals of Chemistry	Distinguish between matter and a substance. (Analyzing)	G09CU01P01	Chemistry and its Basic Terms
	Classify the chemical species from given examples. (Understanding)		
	Identify the representative particles of elements and compounds. (Remembering)		
	Identify and provide examples of different branches of chemistry. (Applying)		
	Differentiate between branches of chemistry. (Understanding)		
	Differentiate among elements, compounds and mixtures. (Remembering)		
	Differentiate among elements, compounds and mixtures. (Remembering)		
	Define atomic number, atomic mass, atomic mass unit. (Remembering)	G09CU01P02	More Terms in Chemistry
	Define relative atomic mass based on C-12 scale. (Remembering)		
	Differentiate between empirical and molecular formula (Understanding)	G09CU01P03	Molecular & Empirical Formulae Types of Ions
	Define ions, molecular ions, formula units and free radicals. (Remembering)		
	Distinguish between atoms and ions. (Analyzing)		
	Differentiate between molecules and molecular ions. (Analyzing)		
	Distinguish between ion and free radical. (Analyzing)		
	Distinguish between ion and free radical. (Analyzing)		
		G09CU01P04	

	Describe how Avogadro's number is related to a mole of any substance. (Understanding)		Avogadro's Number, Moles, and Their Calculations
	Change atomic mass, molecular mass and formula mass into gram atomic mass, gram molecular mass and gram formula mass. Analyzing		
	Relate gram atomic mass, gram molecular mass and gram formula mass to mole. (Applying)		
	Distinguish among the terms gram atomic mass, gram molecular mass and gram formula mass. (Analyzing)		
Structure of Atoms	Describe the contributions that Rutherford made to the development of the atomic theory. (Applying)	G09CU02P01	The History of Human Understanding of Atomic Structure
	Explain how Bohr's atomic theory differed from its. (Analyzing)		
	Describe the structure of an atom including the location of the proton, electron and neutron. (Understanding)	G09CU02P02	Structure of an Atom and Properties of Sub-Atomic Particles
	Define isotopes. (Remembering)		
	Define isotopes. (Remembering)	G09CU02P03	Electronic Configuration of Elements and Isotopes
	Compare isotopes of an atom. (Analyzing)		
	Discuss properties of the isotopes of H, C, Cl, U (Understanding)		
	Draw the structure of different isotopes from mass number and atomic number. (Applying)		
	State the importance and uses of isotopes in various fields of life. (Remembering)		
	Distinguish between shells and sub shells. (Understanding)		
Periodic Table and Periodicity of Properties	Distinguish between a period and a group in the periodic table. (Understanding)	G09CU03P01	Periodic Law, Periodic Table, Shape of the Periodic Table

	State the periodic law. (Remembering)		
	Classify the elements (into two categories: groups and periods) according to the configuration of their outer most electrons. (Analyzing)		
	Determine the demarcation of the periodic table into an s block and p block. (Remembering)		
	Explain the shape of the periodic table. (Analyzing)		
	Recognize the similarity in the chemical and physical properties of elements in the same family of elements. (Understanding)		
	Determine the location of families on the Periodic Table. (Understanding)	G09CU03P02	Families of Elements in Periodic Table
	Recognize the similarity in the chemical and physical properties of elements in the same family of elements. (Understanding)		
	Identify the relationship between electron configuration and the position of an element on the periodic table. (Analyzing)	G09CU03P03	Periodic Trends
	Explain how shielding effect influences periodic trends. (Applying)		
	Describe how electronegativities change within a group and within a period in the periodic table. (Analyzing)		
Structure of Molecules	Find the number of valence electrons in an atom using the Periodic Table. (Applying)	G09CU04P01	Duplet and Octet Rules
	Describe the importance of noble gas electronic configurations. (Understanding)		
	State the octet and duplet rules. (Remembering)		
	Explain how elements attain stability. (Understanding)		
	State the importance of noble gas electronic configurations in the formation of ion. (Applying)		

	Describe the ways in which bonds may be formed. (Remembering)		
	Describe the characteristics of an ionic bond. (Understanding)	G09CU04P02	Ionic Bonding
	Describe the formation of cations from an atom of a metallic element. (Applying)		
	Describe the formation of anions from an atom of a non-metallic element. (Applying)		
	Recognize a compound as having ionic bonds. (Analyzing)		
	Identify characteristics of ionic compounds. (Understanding)		
	Describe the formation of a covalent bond between two non metallic elements. (Understanding)	G09CU04P03	Covalent Bonding
	Describe the formation of a covalent bond between two non metallic elements. (Understanding)		
	Describe with examples single, double and triple covalent bonds. (Understanding)		
	Draw electron cross and dot structures for simple covalent molecules containing single, double and triple covalent bonds. (Applying)		
Physical States of Matter	Compare the physical states of matter with regard to intermolecular forces present between them. (Analyzing)	G09CU05P01	Physical States of Matter Properties of Gases
	Compare the physical states of matter with regard to intermolecular forces present between them. (Analyzing)		
	Effect on the pressure of a gas by a change in the a. volume b. temperature. (Understanding)		

	Explain the properties of gases (diffusion, effusion and pressure). (Understanding)	G09CU05P03	Properties of Liquids and Solids
	Summarize the properties of liquids like evaporation, vapor pressure, boiling point (Understanding)		
	Explain the effect of temperature and external pressure on vapor pressure and boiling point. (Understanding)		
	Describe physical properties of solids (melting and boiling points). (Understanding)		
	Differentiate between amorphous and crystalline solids. (Analyzing)		
	Explain the allotropic forms of solids. (Understanding)		
Solutions	Define the terms: solution, aqueous solution, solute and solvent and give an example of each. (Remembering)	G09CU06P01	Solutions
	Use the rule that "like dissolves like" to predict the solubility of one substance in another. (Understanding)		
	Explain what is meant by the concentration of a solution. (Understanding)		
	Define the terms: solution, aqueous solution, solute and solvent and give an example of each. (Remembering)		
	Define the terms: solution, aqueous solution, solute and solvent and give an example of each. (Remembering)		
	Convert between the Molarity of a solution and its concentration in g/dm ³ . (Applying)		
	Define percentage solution. (Remembering)		
	Define the terms: solution, aqueous solution, solute and solvent and give an example of each. (Remembering)		

	Define the terms: solution, aqueous solution, solute and solvent and give an example of each. (Remembering)	G09CU06P02	More Solutions About
	Convert between the Molarity of a solution and its concentration in g/dm ³ . (Applying)		
	Explain the difference between saturated, unsaturated and supersaturated solutions. (Analyzing)		
	Define percentage solution. (Remembering)		
	Explain the formation of solutions (mixing gases into gases, gases into liquids, gases into solids) and give an example of each. (Understanding)		
	Explain the formation of solutions (mixing liquids into gases, liquids into liquids, liquids into solids) and give an example of each. (Understanding)		
	Explain the formation of solutions (mixing solids into gases, solids into liquids, solids into solids) and give an example of each. (Understanding)		
	Define Molarity. (Remembering)		
	Solve problems involving the Molarity of a solution. (Applying)		
	Describe how to prepare a solution of given Molarity. (Applying)		
	Describe how to prepare dilute solutions from concentrated solutions of known Molarity. (Applying)		
Electrochemistry	List the possible uses of an electrolytic cell. (Understanding)	G09CU07P03	Uses of Electrolytic Cells
	Describe how a battery produces electrical energy. (Understanding)		

	Describe the methods of preparation of alkali metals. (Understanding)		
	Describe the manufacture of sodium metal from fused NaCl. (Understanding)		
	Identify the formation of by products in the manufacture of sodium metal from fused NaCl. (Remembering)		
Chemical Reactivity	Describe the inertness of noble metals. (Understanding)	G09CU08P02	Chemical Reactivity of Elements
	Identify the commercial value of Silver, Gold and Platinum. (Analyzing)		
	Compile some important reactions of halogens. (Applying)		
Electrochemistry	Describe the method of recovering metal from its ore. (Understanding)	G09CU07P04	Electrolytic Refining of Copper and Corrosion
	Explain electrolytic refining of copper. (Understanding)		
	Define corrosion. (Remembering)		
	Describe rusting of iron as an example of corrosion. (Understanding)		
	Summarize the methods used to prevent corrosion. (Understanding)		
	Explain electroplating of metals on steel (using examples of zinc, Tin and chromium plating). (Understanding)		
Structure of Atoms	Write the electronic configurations of the first 18 elements in the Periodic Table. (Remembering)	G09CU02P04-A (partial)	Electronic Configurations
	Write the electronic configurations of the first 18 elements in the Periodic Table. (Remembering)	G09CU02P04-B (partial)	

Physical States of Matter	Explain the properties of liquids and the factors that affect them	G09CU05P02 (partial)	Boyle's Law, Charles' Law and Properties of Liquids
Electrochemistry	Distinguish between electrolytic and voltaic cells. (Analyzing)	G09CU07P02-A (partial) G09CU07P02-B (partial)	Electrolysis and Electrochemical Cells
	Identify the direction of movement of cations and anions towards respective electrodes. (Understanding)		
	Sketch an electrolytic cell, label the cathode and the anode. (Understanding)		
	Describe the nature of electrochemical processes. (Understanding)		
Chemical Reactivity	Explain why alkali metals are not found in the Free State in nature. (Analyzing)	G09CU08P01-A (partial) G09CU08P01-B (partial) G09CU08P01-C (partial)	Chemical Reactivity of Alkali and Alkaline Earth Metals
	Identify elements as an alkali metal or an alkaline earth metal. (Applying)		
	Explain the differences in ionization energies of alkali and alkaline earth metals. (Understanding)		
	Describe the position of sodium in Periodic Table, its simple properties and uses. (Understanding)		
	Describe the position of calcium and magnesium in Periodic Table, their simple properties and uses. (Understanding)		
Electrochemistry	State the common rules used for assigning oxidation numbers to free elements, ions (simple and complex), molecules, atoms. (Remembering)	G09CU07P01	Oxidation and Reduction
	Determine the oxidation number of an atom of any element in a compound. (Applying)		
	Define oxidation state. (Remembering)		
	Define oxidation and reduction in terms of loss or gain of electrons. (Understanding)		

	Define oxidation and reduction in terms of loss or gain of oxygen or hydrogen. (Understanding)		
--	--	--	--

e. Physics

Unit	SLO	Package Code	Package Name
Kinematics	Define the term speed	G09PU02P03	Speed, Distance-Time Graphs and Speed-Time Graphs
	plot and interpret distance-time graph and speed-time graph.		
	determine and interpret the slope of distance-time and speed-time graph.		
	determine from the shape of the graph, the state of a body. i. at rest ii. moving with constant speed iii. moving with variable speed.		
	calculate the area under speed-time graph to determine the distance traveled by the moving body.		
Physical quantities and measurement	explain with examples that Science is based on physical quantities which consist of numerical magnitude and a unit.	G09PU01P02	Physical Quantities, Measuring Instruments, and Rounding
	differentiate between base and derived physical quantities.		
	list the seven units of System International (SI) alongwith their symbols and physical quantities (standard definitions of SI units are not required).		
	interconvert the prefixes and their symbols to indicate multiple and sub-multiple for both base and derived units.		
	identify and explain the limitation of measuring instruments such as metre rule, vernier callipers and screw gauge.		
	describe the need using significant figures for recording and stating results in the laboratory.		
Kinematics	describe using examples how objects can be at rest and in motion simultaneously.	G09PU02P01	Rest and Motion, Scalars and Vectors

	identify different types of motion i.e; translatory, (linear, random, and circular); rotatory and vibratory motions and distinguish among them.		
	differentiate with examples between distance and displacement, speed and velocity.		
	differentiate with examples between scalar and vector quantities.		
	represent vector quantities by drawing.		
Kinematics	define the term speed, velocity and acceleration.	G09PU02P02	Speed, Velocity, and Acceleration
Dynamics	define momentum, force, inertia, friction, centripetal force.	G09PU03P01	Force and Momentum
Dynamics			
The turning effect of forces	define like and unlike parallel forces.		
The turning effect of forces	state head to tail rule of vector addition of forces/vectors.		
Dynamics	solve problem using the equation $\text{Force} = \text{change in momentum} / \text{change in time}$.		
Dynamics	state Newton's laws of motion.	G09PU03P02	Newton's Laws of Motion and Mass and Weight
	distinguish between mass and weight and solve problem using $F = ma$, and $w = mg$.		
Dynamics	explain the effect of friction on the motion of a vehicle in the context of tyre surface, road conditions including skidding, braking force.	G09PU03P04	Friction and Circular Motion

	demonstrate that rolling friction is much lesser than sliding friction.		
	list various methods to reduce friction.		
	explain that motion in a curved path is due to a perpendicular force on a body than changes direction of motion but not speed.		
The turning effect of forces	describe how a force is resolved into its perpendicular components.	G09PU04P01	Moment And Resolution of Forces
	determine the magnitude and direction of a force from its perpendicular components.		
	define moment of force or torque as moment = force x perpendicular distance from pivot to the line of action of force.		
	explain the turning effect of force by relating it to everyday life.		
	state the principle of moments.		
	solve problems on simple balanced systems when bodies are supported by one pivot only.		
Gravitation	state Newton's law of gravitation.	G09PU05P01	Gravitation
	explain that the gravitational forces are consistent with Newton's third law.		
	explain gravitational field as an example of field of force.		
	define weight (as the force on an object due to a gravitational field.)		
	discuss the importance of Newton's law of gravitation in understanding the motion of satellites.		
	solve problems using Newton's law of gravitation.		
Work and energy	define work and its SI unit.	G09PU06P01	Work and Energy
	calculate work done using equation Work = force x distance moved in the direction of force		
	define energy, kinetic energy and potential energy. State unit of energy.		
	prove that Kinetic Energy $E_k = \frac{1}{2}mv^2$ and potential energy $E_p = mgh$ and solve problems using these equations.		

	list the different forms of energy with examples.		
	describe the processes by which energy is converted from one form to another with reference to o fossil fuel energy o hydroelectric generation o solar energy o nuclear energy o geothermal energy o wind energy o biomass energy		
Work and energy		G09PU06P02	Electricity Generation And $E=mc^2$
	state mass energy equation $E = mc^2$ and solve problems using it. describe the process of electricity generation by drawing a block diagram of the process from fossil fuel input to electricity output.		
Work and energy	Revision Video	G09PU06P03	Sources of Energy And Efficiency
	list the environmental issues associated with power generation.		
	differentiate energy sources as non renewable and renewable energy sources with examples of each.		
	explain by drawing energy flow diagrams through steady state systems such as Filament lamp, a power station, a vehicle traveling at a constant speed on a level road.		
	define efficiency of a working system and calculate the efficiency of an energy conversion using the formula o $\text{efficiency} = \frac{\text{energy converted into the required form}}{\text{total energy input}}$		
	explain why a system cannot have an efficiency of 100%.		
Properties of matter	define the term 'Density'	G09PU07P02	Density and Pressure
	compare the densities of a few solids, liquids and gases.		
	define the term pressure (as a force acting normally on unit area).		
	explain how pressure varies with force and area in the context of everyday examples.		
	explain that the atmosphere exerts a pressure.		
	describe how the height of a liquid column may be used to measure the atmospheric pressure.		

	describe that atmospheric pressure decreases with the increase in height above the Earth's surface.		
Properties of matter	state Pascal's law.	G09PU07P03	Pascal's Law and Archimedes' Principle
	apply and demonstrate the use with examples of Pascal's law		
	state relation for pressure beneath a liquid surface to depth and to density i.e., $p = \rho gh$ and solve problems using this equation.		
	state Archimedes principle.		
	determine the density of an object using Archimedes principle.		
	state the upthrust exerted by a liquid on a body.		
	state principle of floatation.		
Thermal properties of matter	define temperature (as quantity which determine the direction of flow of thermal energy).	G09PU08P01	Heat and Heat Capacity
	define heat (as the energy transferred resulting from the temperature difference between two objects).		
	describe rise in temperature of a body in term of an increase in its internal energy.		
	define the terms heat capacity and specific heat capacity.		
	describe heat of fusion and heat of vaporization (as energy transfer without a change of temperature for change of state).		
	solve numerical problems based on the mathematical relations learnt in this unit		
	describe experiments to determine heat of fusion and heat of vaporization of ice and water respectively by sketching temperature-time graph on heating ice.		
Thermal properties of matter	explain the process of evaporation and the difference between boiling and evaporation.	G09PU08P02	Evaporation and Thermal Expansion
	explain that evaporation causes cooling.		
	list the factors which influence surface evaporation.		
	describe qualitatively the thermal expansion of solids (linear and volumetric expansion).		

	explain the thermal expansion of liquids (real and apparent expansion) list basic thermometric properties for a material to construct a thermometer.		
Transfer of heat	recall that thermal energy is transferred from a region of higher temperature to a region of lower temperature. describe in terms of molecules and electrons, how heat transfer occurs in solids. write examples of good and bad conductors of heat and describe their uses. explain the convection currents in fluids due to difference in density. state some examples of heat transfer by convection in everyday life. explain insulation reduces energy transfer by conduction. describe the process of radiation from all objects. explain that energy transfer of a body by radiation does not require a material medium and rate of energy transfer is affected by: - o Colour and texture of the surface - o Surface temperature - o Surface area	G09PU09P01	Transfer of Heat
<i>Physical quantities and measurement</i>	write the answer in scientific notation in measurements and calculations.	G09PU01P01-B(partial)	Scientific Notation
<i>Physical quantities and measurement</i>	describe the working of vernier callipers and screw gauge for measuring length.	G09PU01P03-A(partial) G09PU01P03-B(partial)	Measuring Instruments
<i>Kinematics</i>	solve problems related to freely falling bodies using 10 ms^{-2} as the acceleration due to gravity.	G09PU02P04-C(partial)	Free-Fall Problems with 10 ms^{-2} Gravity
Dynamics	state the law of conservation of momentum.	G09PU03P03-A(partial)	Momentum Conservation: Theory and Problems
Dynamics	use the principle of conservation of momentum in the collision of two objects.		
Dynamics	determine the velocity after collision of two objects using the law of conservation of momentum.		
<i>Properties of matter</i>	explain that a force may produce a change in size and shape of a body.	G09PU07P04-A(partial) G09PU07P04-B(partial)	

<i>Properties of matter</i>	define the terms Stress, Strain and Young's modulus.		Force Impact on Elasticity, Stress, and Strain
<i>Properties of matter</i>	state Hooke's law and explain elastic limit.		
Dynamics	calculate centripetal force on a body moving in a circle using mv^2/r .	G09PU03P05-A(partial)	Centripetal Force in Circular Motion

5. How to Use This Document in Class

This guide is designed to make lesson planning and classroom delivery easier. Teachers can use it in the following ways:

- **Lesson Planning:** Before each lesson, check the tables to see which video(s) cover the relevant SLO(s) for your class.
- **Classroom Instruction:** Play the appropriate video during class to illustrate concepts, reinforce learning, or introduce new topics.
- **Curriculum Alignment:** Ensure videos are paired with textbook units to maintain alignment with the Pakistan National Curriculum.
- **Revision and Reinforcement:** Use videos to review previous lessons or provide extra practice for students who need it.
- **Multi-Grade or Mixed-Level Classes:** Teachers can identify videos suitable for different grade levels and adjust classroom activities accordingly.
- **Assessment:** Each video package has a corresponding assessment, named with the same package code, which can be found in the accompanying Excel sheets. Teachers can use these assessments to check students' understanding immediately after showing the video or during lesson follow-ups.

The document is structured to make it easy to locate videos quickly, so you spend more time teaching and less time searching.

6. Ethical and Copyright Considerations

All digital content provided through this platform is procured by the Ministry of Federal Education and Professional Training and aligns with the Pakistan National Curriculum. When using these resources, teachers should keep the following points in mind:

- **Authorized Use Only:** Videos are for educational purposes within classrooms or official teaching activities. Do not distribute, copy, or upload content to public platforms without permission.
- **Attribution:** When using content in presentations or shared materials, credit the Ministry as the source.
- **Respect for Learners:** Ensure all content is used in a way that supports inclusive and respectful learning environments.
- **No Modification Without Approval:** Do not alter video content, subtitles, or audio tracks unless officially authorized.
- **Data Privacy:** Avoid sharing student information or recordings when using videos in digital or remote learning settings.

Following these guidelines ensures that teachers use the digital content responsibly while maintaining legal and ethical standards.

7. Conclusion

This guide is designed to make using digital learning videos simple, efficient, and aligned with the Pakistan National Curriculum and Student Learning Outcomes. By following the mapping tables and video codes, teachers can quickly identify the right resources for each lesson, support student learning effectively, and plan classroom activities with confidence.

We encourage teachers to explore the videos, pair them with assessments, and integrate them into lesson plans to create engaging and interactive learning experiences.

For any questions, technical issues, or additional support, please contact the Federal Directorate of Education (FDE)'s content team or the designated focal points.

Empower your students, explore the digital resources, and make every lesson an engaging step toward learning success.